

OSLO BLINDERN, Norway

WMO: 014920

Lat: 59.9422N

Lon: 10.7200E

Elev: 97

StdP: 100.17

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.9	-11.7	-16.8	0.9	-12.2	-14.8	1.0	-9.4	8.9	0.5	8.0	1.0	1.7	30	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.4	26.9	17.3	25.0	16.7	23.2	15.7	18.7	24.1	17.8	22.8	16.9	21.6	2.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	12.1	20.3	15.9	11.4	19.4	15.0	10.8	18.7	53.2	24.2	50.4	22.8	47.9	21.6	21.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.9	6.8	6.0	-16.1	29.2	2.4	2.6	-17.9	31.1	-19.3	32.6	-20.7	34.1	-22.4	36.0		
			-16.6	19.8	2.4	1.0	-18.3	20.5	-19.7	21.0	-21.0	21.6	-22.7	22.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	-2.5	-1.8	1.4	6.4	11.5	15.3	18.0	16.8	12.5	6.6	2.1	-1.5
	DBStd	8.11	4.55	4.32	3.92	3.32	3.76	2.81	2.75	2.63	2.79	3.28	3.71	4.63
	HDD10.0	1846	389	331	268	117	25	1	0	0	9	113	236	358
	HDD18.3	4152	647	564	526	358	215	98	41	64	174	363	486	616
	CDD10.0	791	0	0	1	8	72	160	247	211	85	8	0	0
	CDD18.3	56	0	0	0	0	3	8	29	16	0	0	0	0
	CDH23.3	377	0	0	0	0	24	77	200	76	1	0	0	0
	CDH26.7	66	0	0	0	0	3	13	43	7	0	0	0	0
Wind	WSAvg	2.7	2.6	2.5	2.7	2.9	3.0	2.9	2.6	2.6	2.7	2.6	2.8	2.5
Precipitation	PrecAvg	772	52	37	45	43	52	69	81	87	88	84	77	53
	PrecMax	1173	139	89	121	91	116	143	181	228	229	188	279	137
	PrecMin	561	4	1	0	0	1	17	14	12	28	12	11	7
	PrecStd	129	32	22	31	25	28	32	37	51	41	47	50	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.3	8.2	14.9	19.6	26.3	28.6	30.4	27.5	22.2	15.5	11.6	8.9
		MCWB	5.0	4.0	7.8	10.8	15.4	16.4	18.6	18.1	15.6	11.7	9.7	7.6
	2%	DB	5.8	6.0	11.4	16.9	22.9	25.6	27.7	25.5	20.1	13.7	10.1	7.5
		MCWB	4.1	3.4	5.9	9.4	13.6	15.4	18.0	17.6	14.4	11.1	8.7	6.3
	5%	DB	4.5	4.8	8.7	14.5	20.6	23.1	25.5	23.6	18.5	12.7	8.7	6.0
		MCWB	3.0	3.0	4.0	8.1	12.9	14.6	17.4	16.7	13.7	10.6	7.3	4.8
	10%	DB	3.3	3.6	6.6	12.3	18.3	21.0	23.5	21.7	17.1	11.6	7.4	4.6
		MCWB	2.0	2.2	3.0	6.9	11.8	13.4	16.3	15.7	13.1	9.8	6.2	3.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.8	5.5	8.4	11.9	17.0	18.0	20.1	19.6	17.1	13.2	10.5	8.0
		MCDB	6.4	7.0	14.2	17.8	23.6	24.6	26.5	24.1	19.8	14.2	11.1	8.6
	2%	WB	4.6	4.3	6.2	10.0	15.1	16.8	19.0	18.5	15.8	12.1	8.9	6.5
		MCDB	5.4	5.3	10.3	14.9	21.2	22.8	25.1	23.1	18.4	13.1	9.8	7.3
	5%	WB	3.4	3.4	4.9	8.7	13.7	15.6	18.1	17.6	14.8	11.1	7.6	5.0
		MCDB	4.2	4.4	7.5	12.6	18.7	21.2	23.5	22.0	17.2	12.3	8.4	5.8
	10%	WB	2.2	2.3	3.8	7.6	12.5	14.6	17.3	16.7	13.9	10.1	6.4	3.5
		MCDB	3.1	3.4	6.0	11.3	17.5	19.3	22.3	20.6	16.2	11.4	7.3	4.4
Mean Daily Temperature Range	5% DB	MDBR	4.4	5.2	7.2	8.1	8.9	8.7	8.4	7.9	7.3	5.6	4.2	4.3
		MCDBR	5.1	6.3	10.5	12.0	12.4	12.4	11.8	10.4	9.4	6.9	5.9	5.4
		MCWBR	4.4	4.5	6.1	6.4	6.0	5.5	5.0	4.5	5.2	4.8	5.3	5.0
	5% WB	MCDBR	4.5	5.1	9.4	10.1	10.6	10.3	10.2	8.9	7.7	5.4	5.4	5.0
		MCWBR	4.3	4.1	5.9	5.7	5.6	5.2	4.8	4.4	4.9	4.4	5.1	4.9
Clear-Sky Solar Irradiance	taub	0.260	0.302	0.328	0.371	0.367	0.356	0.377	0.370	0.338	0.324	0.280	0.233	
	taud	2.207	2.287	2.333	2.317	2.402	2.452	2.413	2.428	2.508	2.455	2.291	2.122	
	Ebn at Noon	528	703	803	827	861	876	845	822	795	679	502	405	
	Edh at Noon	41	67	87	106	105	101	103	94	74	57	38	28	
All-Sky Solar Radiation	RadAvg	0.29	0.89	2.38	3.77	4.89	5.60	5.00	3.94	2.61	1.15	0.38	0.19	
	RadStd	0.05	0.16	0.25	0.51	0.50	0.42	0.65	0.31	0.30	0.22	0.10	0.02	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (58 neighbors)	N/A	N/A	+1.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page